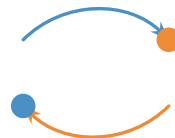


Chapter 3

Interdependence and the Gains from Trade

Chapter 1 told us that trade can make everyone better off. This chapter asks why that is true.

We will build a small model of two countries and two goods, and use it to answer a real puzzle: why does a country ever trade with a partner who is worse at making everything?



Interdependence

We rely on people all over the world whom we will never meet to provide us with food, clothing, electronics, and countless other goods, and in return we provide something they want.

Almost nobody today is self-sufficient.

This chapter asks why people, and nations, choose to depend on one another this way, and how that dependence pays off.

Setting Up an Example

To see where the gains from trade come from, we will build the simplest possible model: two countries, Taiwan and Cambodia, each producing two goods, chip batches and tons of rice, using only one input, labor.

To keep the model simple, think of each country as a single representative worker who can work 8 hours a day, and who can switch between the two goods at a constant rate.

Example. Taiwan needs 20 minutes of labor to produce one chip batch and 10 minutes to produce one ton of rice. Cambodia needs 60 minutes to produce one chip batch and 15 minutes to produce one ton of rice.

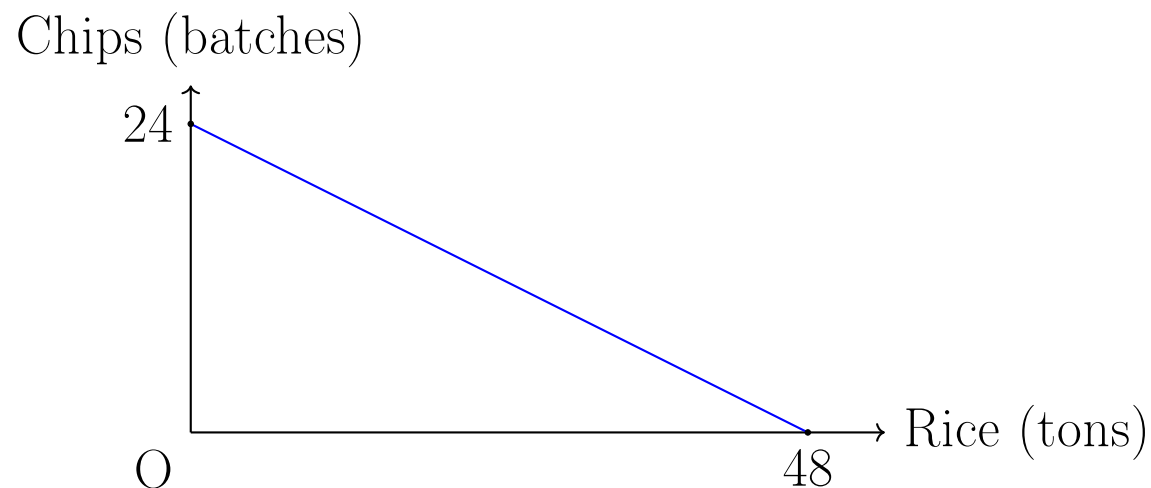
Labor Requirements

	Minutes per chip batch	Minutes per ton of rice
Taiwan	20	10
Cambodia	60	15

This table records the same two numbers from the previous slide, laid out the way we will use them throughout the chapter: minutes of labor needed per unit, for each country and each good.

With 8 hours, 480 minutes, of labor a day, this is exactly the information needed to draw each country's production possibilities frontier.

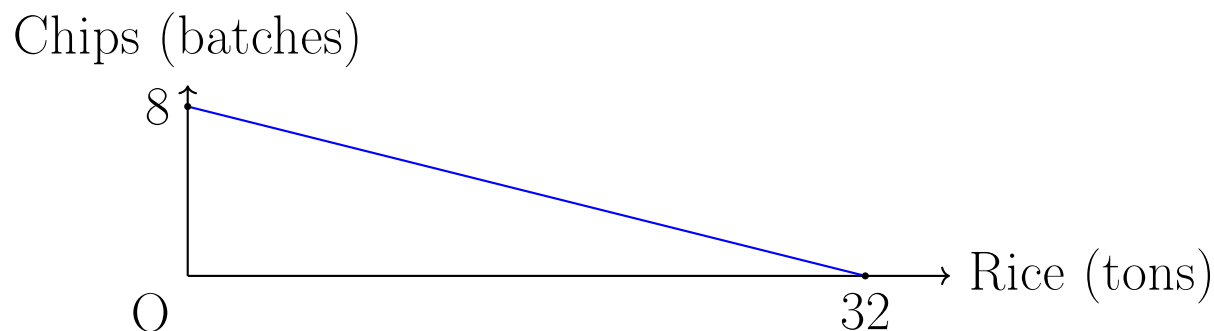
Taiwan's Production Possibilities



If Taiwan spends all 8 hours making chips, it can produce 24 chip batches and no rice. If it spends all 8 hours on rice, it can produce 48 tons and no chips.

Any combination in between is also possible, and the frontier is a straight line because Taiwan's technology lets it switch between the two goods at a constant rate.

Cambodia's Production Possibilities



Cambodia can produce at most 8 chip batches or 32 tons of rice a day.

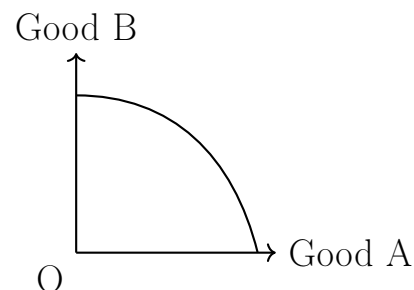
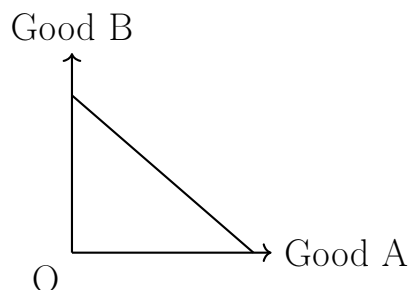
Cambodia needs more labor than Taiwan for either good, so its frontier sits entirely inside Taiwan's: Taiwan can simply produce more of everything.

A Note on the Shape of the PPF

Taiwan's and Cambodia's frontiers are straight lines because we assumed each country can switch between rice and chips at a constant rate.

Remark. Real economies rarely satisfy this assumption: workers, land, and machines each suit one good better than the other, so shifting more resources toward a good gets progressively less efficient. Most real production possibilities frontiers bow outward instead of running straight, reflecting an opportunity cost that rises the more of a good is produced.

Constant opportunity cost Increasing opportunity cost



Life without Trade

Suppose that, without trade, each country splits its 8 hours evenly between the two goods.

Example. Taiwan spends 4 hours on each good and produces and consumes 12 chip batches and 24 tons of rice a day. Cambodia spends 4 hours on each good and produces and consumes 4 chip batches and 16 tons of rice a day.

Without trade, each country's production possibilities frontier is also its consumption possibilities frontier: a country can only consume what it makes for itself.

Where Do the Gains from Trade Come From?

Definition. **Absolute advantage** is the ability to produce a good using fewer inputs than another producer.

Taiwan needs less labor than Cambodia to produce either good, 20 minutes versus 60 for a chip batch, and 10 minutes versus 15 for a ton of rice, so Taiwan has an absolute advantage in both goods.

If absolute advantage were what determined trade, Taiwan would have no reason to trade with Cambodia at all, yet as we will see, both countries can still gain.

A Second Measure of Cost

Absolute advantage compares the inputs two producers need. A second measure compares what each producer gives up.

Definition. The **opportunity cost** of a chip batch is the amount of rice that could have been produced with the labor used to produce it instead.

Since Taiwan needs 20 minutes for a chip batch and 10 minutes for a ton of rice, the 20 minutes spent on one chip batch could instead have produced 2 tons of rice.

Taiwan's opportunity cost of a chip batch is therefore 2 tons of rice.

Comparing Opportunity Costs

	Opp. cost of 1 chip batch	Opp. cost of 1 ton rice
Taiwan	2 tons of rice	0.5 chip batch
Cambodia	4 tons of rice	0.25 chip batch

Taiwan gives up less rice to produce a chip batch than Cambodia does, 2 tons versus 4.

Cambodia gives up less in chip batches to produce a ton of rice than Taiwan does, 0.25 versus 0.5.

Each country is the lower-cost producer of a different good.

Comparative Advantage

Definition. **Comparative advantage** is the ability to produce a good at a lower opportunity cost than another producer.

Taiwan has the comparative advantage in chips, since its opportunity cost of a chip batch, 2 tons of rice, is lower than Cambodia's, 4 tons.

Cambodia has the comparative advantage in rice, since its opportunity cost of a ton of rice, 0.25 chip batch, is lower than Taiwan's, 0.5 chip batch.

Remark. It is impossible for one country to have a comparative advantage in both goods. Since the opportunity cost of one good is the inverse of the opportunity cost of the other, a country that is relatively cheap at producing one good is automatically relatively expensive at producing the other.

Specializing and Trading

Suppose Taiwan shifts more of its time toward chips, producing 18 chip batches and 12 tons of rice a day, and Cambodia specializes completely in rice, producing 32 tons and no chips.

Cambodia then trades 15 tons of rice to Taiwan in exchange for 5 chip batches, a price of 3 tons of rice per chip batch.

That price is no accident: 3 lies between Taiwan's opportunity cost of a chip batch, 2 tons of rice, and Cambodia's, 4 tons, which is exactly the range in which both countries are willing to trade.

Imports and Exports

Definition. **Imports** are goods produced abroad and sold domestically.

Definition. **Exports** are goods produced domestically and sold abroad.

In the trade we just described, Taiwan exports chip batches and imports rice; Cambodia exports rice and imports chip batches.

Every export for one country is somebody else's import: the 15 tons of rice Cambodia exports are exactly the 15 tons of rice Taiwan imports.

The Gains from Trade

	Chip batches	Tons of rice
Taiwan, without trade	12	24
Taiwan, with trade	13	27
Cambodia, without trade	4	16
Cambodia, with trade	5	17

After trade, Taiwan consumes one more chip batch and three more tons of rice than it could produce alone, and Cambodia consumes one more of each good than it could produce alone.

Both countries end up with more of both goods than they had access to before, without anyone working any additional hours.

Resolving the Puzzle

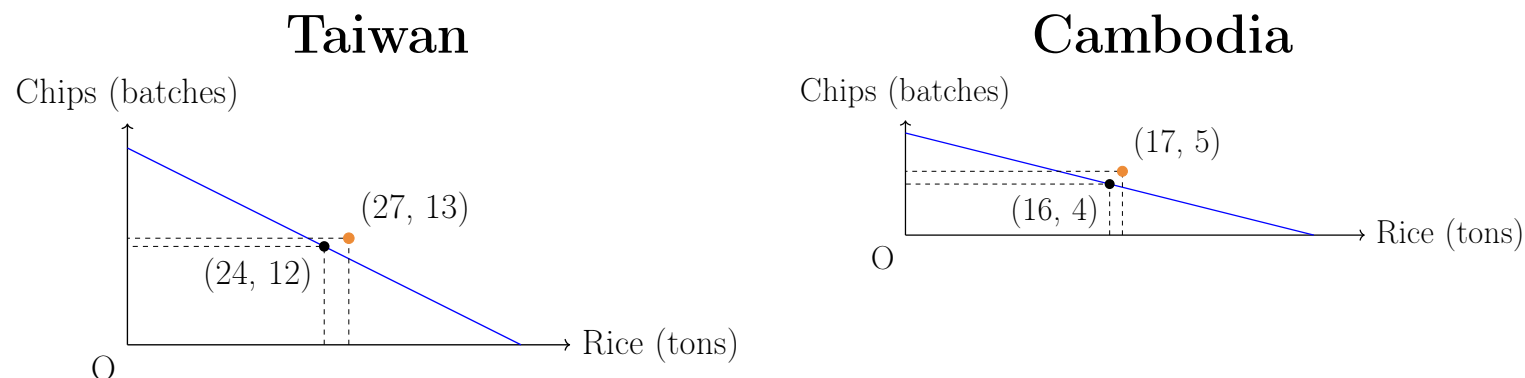
Taiwan has an absolute advantage in both goods, yet it still gains from trading with a partner who is worse at making everything.

The reason is that trade is governed by comparative advantage, not absolute advantage.

By each country specializing in the good it produces at the lower opportunity cost, chips for Taiwan and rice for Cambodia, total production of both goods rises, and both countries can share in that larger total.

Remark. Trade lets each country consume outside its own production possibilities frontier, a combination of goods it could never have produced by itself.

Example: Consuming Outside the PPF



- without trade (on the frontier)
- with trade (outside the frontier)

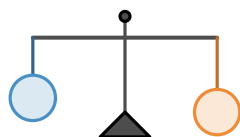
Each country's darker point sits exactly on its own PPF, the most it could produce and consume alone. The orange point, the combination it actually consumes with trade, lies outside that frontier on both graphs.

Example: The Surgeon and Her Assistant

The same logic applies to individual people, not just countries.

A surgeon who could also type faster than anyone in her office should still hire an assistant to handle her scheduling and paperwork, since every hour she spends on typing is an hour she is not spending on surgery, and her opportunity cost of an hour is far higher than her assistant's.

She has an absolute advantage at both tasks, but a comparative advantage only at surgery.



Should Countries Trade with One Another?

Definition. **Imports** are goods produced abroad and sold domestically. **Exports** are goods produced domestically and sold abroad.

The principle of comparative advantage says a country should produce, and export, the good for which it has the lower opportunity cost, and import the good for which its opportunity cost is higher.

This is exactly why Taiwan exports semiconductors, the good it can produce at the lowest opportunity cost, while importing food and raw materials that other countries can produce more cheaply in opportunity-cost terms.

Discussion: Who Has the Advantage?

In an hour, a worker in Country A can assemble 4 bicycles or 2 refrigerators. In the same hour, a worker in Country B can assemble 1 bicycle or 1 refrigerator.

1. Which country has an absolute advantage in bicycles? In refrigerators?
2. What is each country's opportunity cost of one bicycle, in refrigerators?
3. Which country has a comparative advantage in bicycles? In refrigerators?
4. If the two countries trade, which good should Country A export?

Answer: Who Has the Advantage?

1. Country A has an absolute advantage in both goods, since its worker produces more of each in an hour.
2. Country A's opportunity cost of a bicycle is 0.5 refrigerator (4 bicycles cost it 2 refrigerators), while Country B's is 1 refrigerator.
3. Country B has the lower opportunity cost of a bicycle, so Country B has the comparative advantage in bicycles, and Country A has the comparative advantage in refrigerators.
4. Country A should export refrigerators and import bicycles, even though it is the more productive country overall.

The Legacy of Adam Smith and David Ricardo

Adam Smith argued in 1776 that a family, like a tailor who buys shoes rather than making them, gains by specializing in what it does best and trading for the rest.

In 1817, the economist David Ricardo formalized this insight as the principle of comparative advantage, using the example of England and Portugal trading cloth and wine.

Their conclusions still anchor the case for free trade among economists today, even though economists disagree sharply about many other policy questions.

Example: Taiwan's Rice Import Quota

Taiwan is a highly efficient producer of rice, but rice farming still uses land and labor that could otherwise go toward higher-value activities.

As part of joining the World Trade Organization, Taiwan agreed to allow a minimum quota of rice imports each year rather than blocking foreign rice entirely, a policy that lowers costs for rice buyers but is also strongly opposed by domestic rice farmers, who compete directly with the imports.

This tension is exactly what the principle of comparative advantage predicts: freer trade tends to help consumers and the industries that export, while it can hurt the specific producers who compete with imports, even as it raises the country's total economic pie.

Example: U.S. Tariffs on China

Starting in 2025, the United States and China raised tariffs on each other's goods several times, pushing combined U.S. duties on many Chinese goods above 50%.

At a summit in Busan, South Korea that October, Presidents Trump and Xi agreed to a one-year truce: the U.S. tariff fell from 57% to 47%, and China paused new export controls on rare earth minerals and increased its purchases of American soybeans.

As of September 2026, the two governments are negotiating further cuts ahead of another summit. Tariffs and export controls are bargaining leverage here, and both countries pay a cost while the dispute continues.

Example: The 2026 U.S.–Taiwan Trade Agreement

In February 2026, Taiwan and the United States signed a reciprocal trade agreement. The United States set a 15% tariff on imports from Taiwan, matching its rate on Japan, South Korea, and the EU.

In exchange, Taiwan agreed to eliminate or reduce tariffs on 99% of its tariff lines (product categories) covering U.S. industrial and agricultural goods. Its tariffs of up to 26% on U.S. beef, dairy, and corn, and its 17.5% tariff on U.S. passenger cars, both fell to zero immediately.

Some politically sensitive tariffs fell only partway: Taiwan's tariffs on U.S. pork belly (40%) and ham (32%) both fell to just 10%, protecting Taiwanese pork farmers who compete directly with U.S. imports. Taiwan also committed to billions of dollars in U.S. energy, aircraft, and grid-equipment purchases through 2029.

Think-Pair-Share: Are Trade Restrictions Ever Justified?

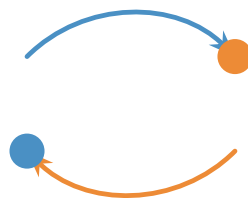
Consider the examples above: Taiwan's rice import quota, the U.S.-China tariff dispute, and Taiwan's own recent tariff cuts on U.S. autos and agriculture, but not (yet) on pork.

- In each case, who specifically gains from the trade restriction (or from removing it), and who specifically loses?
- Is it possible for a policy to help one industry while making the country's total economic pie smaller?
- How would we explain the difference between “some people gain” and “the country gains” to a friend who has not taken this course?

Chapter in a Nutshell: Interdependence

Trade and interdependence are desirable because they let everyone enjoy a greater quantity and variety of goods and services than self-sufficiency ever could.

- **Absolute advantage** measures which producer needs fewer inputs to make a good.
- **Comparative advantage** measures which producer gives up less of other goods to make it, and it is comparative advantage, not absolute advantage, that determines who should produce what.



Chapter in a Nutshell: Specialization and Trade

When each producer specializes in the good in which they hold a comparative advantage, and then trades with others, total production rises and everyone involved can consume more than they could have produced alone.

This logic applies equally to two people sharing chores, two firms in the same industry, and two countries on opposite sides of the world, which is why economists apply the same principle of comparative advantage to argue for free trade among nations.

Key Concepts

- Absolute advantage and comparative advantage are the two ways to compare the cost of producing a good across two producers.
- Opportunity cost is the tool used to measure comparative advantage, and it is always the inverse across the two goods a producer can make.
- Imports and exports describe the flow of goods that specialization and trade create between countries.
- The gains from trade come from comparative advantage, which is why a country can gain from trading with a partner who is worse at producing everything.